

Epilogue

Now you have another perspective to employ in your efforts to prevent and rehabilitate back disorders. I hope that from now on, when you read that painful backs just happen, or that they cannot be diagnosed to guide therapy, or that they are solely the function of psychological factors or the compensation system, you will pause and consider that you are reading the musings of those who have reached the limit of their underdeveloped expertise.

Not that anyone has all the answers. With each experiment we perform, we may obtain some new insight, but generally we are confronted with many more questions. So, with each experiment we become relatively more aware of what we don't know.

Consider the viewpoint of this text and blend it with your own clinical wisdom and experiences. The very best clinicians with whom I have had the pleasure of working had wonderful clinical skills and insights, but also a solid scientific foundation.

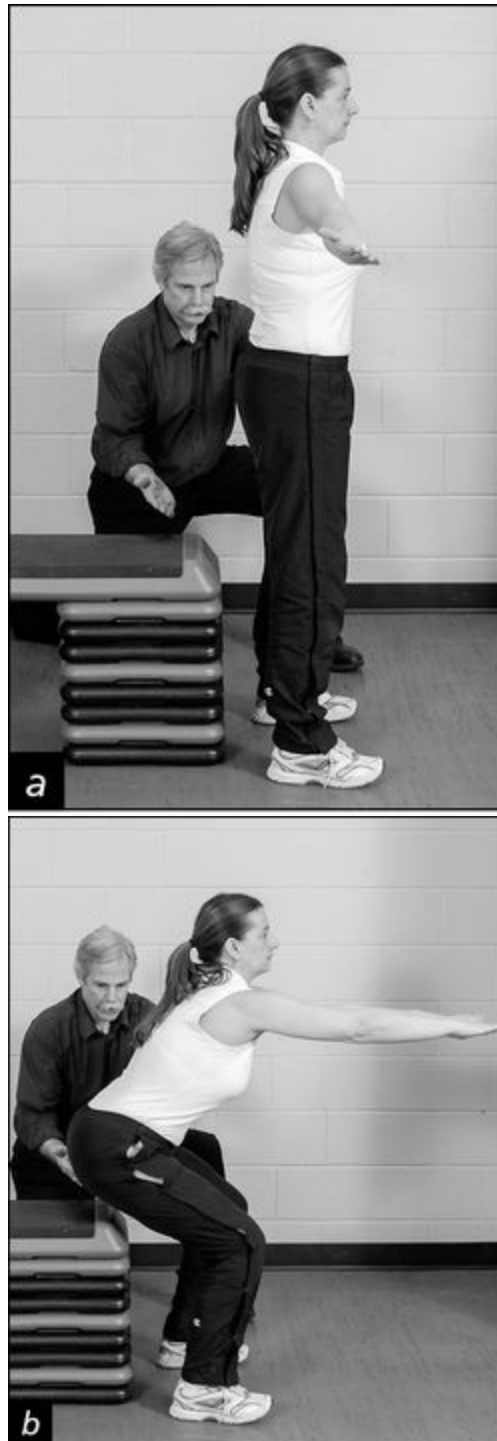
To the clinicians—I wish the confidence to continue with what you know works and the inspiration and leadership to try new approaches when things are problematic. To the students of clinical disciplines and those involved in the continuing development of the scientific foundation—there are so many more exciting experiments to perform on this fascinating part of our human anatomy, the low back. May we all enjoy the continuing journey.

Handouts for Patients or Clients

With great reservation I present these handouts of cues to use when prescribing exercises for patients. As I have mentioned, handing out pamphlets showing back exercises ensures only mediocre practice. To avoid such practice and to use these handouts effectively, you must select appropriately designed exercises for a patient only after you have carefully assessed that patient. Further, you must fine-tune each exercise for the patient by working with him or her to adjust posture, motion, and motor patterns to eliminate pain. This tuning is essential for making the exercise most tolerable and beneficial. During the workshopping of each exercise, you identify problems or limitations and follow up by designing and trying appropriate corrections with the patient. You may reproduce each handout for use with your clients, but you may also access blank copies of these handouts in the web resource available at www.HumanKinetics.com/LowBackDisorders. The handouts in the web resource include fields in which you may type customized cues and steps for each patient as well as his or her unique dosage. Not everyone will do every exercise. Exercises progress at various rates. Add new exercises only one at a time.

Success at this stage will elevate you to the elite level of practice—but you need to consider all principles in this book. First, work to remove all exacerbators of the pain. Then consider and evaluate the patient's tolerance and capabilities and stay within those parameters, avoid injury mechanisms, and correct perturbed patterns.

Potty Squat



Example Cues

1. Stand with feet shoulder-width apart, arms to sides, and hands supinated.
2. Sniff air and appropriately stiffen the torso.
3. Squat the pelvis back, controlling the knees so they do not drift forward beyond the toes. The arms move forward during this descent and the hands roll over into pronation.
4. To begin the ascent, “spread the floor,” cueing the gluteal muscles.
5. Think of pulling the hips forward to encourage the hip hinge.
6. During the ascent, extend the arms while supinating the hands.

From S. McGill, 2016, *Low back disorders*, 3rd ed. (Champaign, IL: Human Kinetics).

Flossing, Beginner





Example Cues

1. If a bulging disc is the cause of radiating neural symptoms, begin with the appropriate posture to reduce the bulge.
2. Find a sitting posture that minimizes pain and maximizes the reflexes.
3. Floss one leg and the head and neck 10 cycles, never pushing to the range that triggers pain or symptoms.
4. Repeat with the other leg and assess reaction immediately after the

session and the following day.

5. If symptoms are not worse, continue the 10 reps per leg 2 or 3 times per day.
6. If symptoms worsen, stop immediately.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Cat–Camel



Example Cues

1. Separate knees to free up the hips.
2. Cycle between flexion and extension of the entire spine (cervical, thoracic, lumbar) no more than 8 reps.
3. Do not push the end range because this is not a stretch. It's the motion that is therapeutic.
4. If you wish, add lateral bending motion cycles.

From S. McGill, 2016, *Low back disorders*, 3rd ed. (Champaign, IL: Human Kinetics).

Curl-Up, Beginner

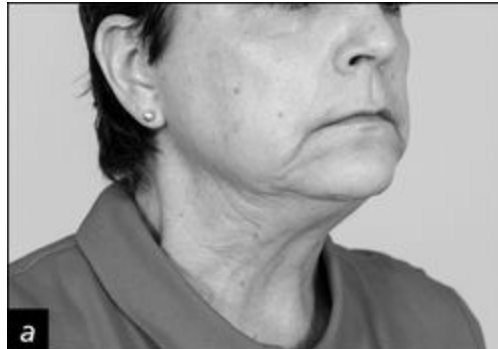


Example Cue

Mildly contract the abdominal muscles and the lifting effort to ensure there is no pain.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Isometric Exercises for the Neck





Example Cues

1. Position the head and neck in a neutral posture.
2. Maintaining this posture, bring the fists underneath the chin.
3. Push tongue to roof of mouth behind front teeth.
4. Push up with the fists, matching the resistance with the neck muscles; do not allow motion.
5. Note that just the exercise shown in a may be sufficient.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Curl-Up, Intermediate



Example Cues

1. Prebrace the abdominal muscles.
2. Lift elbows.
3. Raise head and shoulders slightly.
4. Hold 8 seconds (2 breaths).
5. Repeat 6 times (set 1), rest 20 seconds, repeat 4 reps (set 2), rest, repeat 2 reps (set 3).

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Curl-Up, Advanced

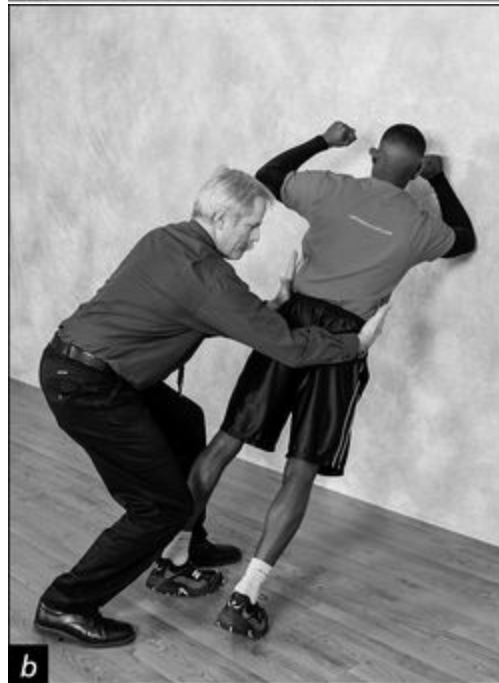
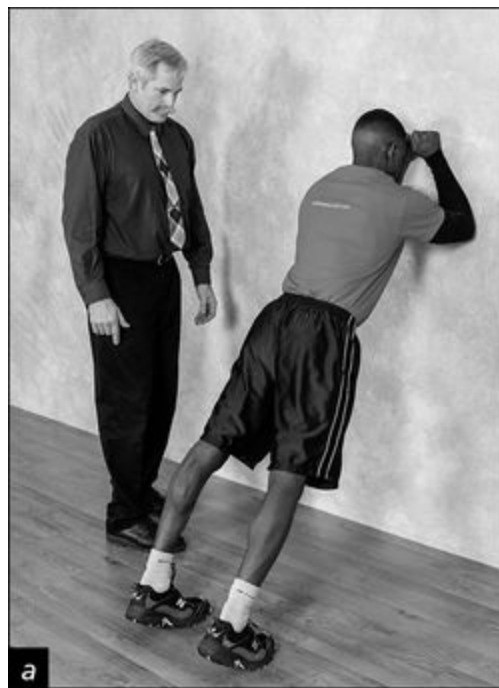


Example Cues

1. Push tongue to roof of mouth.
2. Add full-tide breathing while maintaining the appropriate abdominal brace.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Remedial Side Bridge for Deconditioned, Chronic Patient

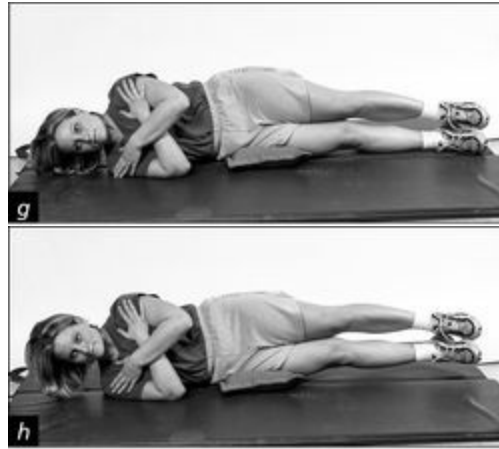


Example Cues

1. The challenge of effort and pain control begins with adjusting the distance from the feet to the wall.
2. Keep the spine aligned in the neutral curves but allow slight variations to reduce pain.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Remedial Side Bridge for Painful Shoulders 1



Example Cues

1. Adjusting a folded towel under the hip or between the waist and floor may increase the pain tolerance.
2. Lift the legs from the floor and no further.
3. Prebrace the torso appropriately.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Remedial Side Bridge for Painful Shoulders 2





Follow the same cues as other versions of the remedial side bridge.
From S. McGill, 2016, *Low back disorders*, 3rd ed. (Champaign, IL: Human Kinetics).

Side Bridge, Beginner



Example Cues

1. Assume a neutral spine in the starting posture by adjusting the pelvis in the frontal plane.
2. To pull up into the bridge posture, extend through the hips in a squat motion.
3. Pull down with the hand, cupping the shoulder and pulling the elbow across the chest.
4. Tune the stiffening of the torso with mental focus on the latissimus dorsi and abdominal wall.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Side Bridge, Intermediate



Example Cue

Think about neutral spine postures: “Pull the hips through” is the most common cue.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Side Bridge, Intermediate Variation

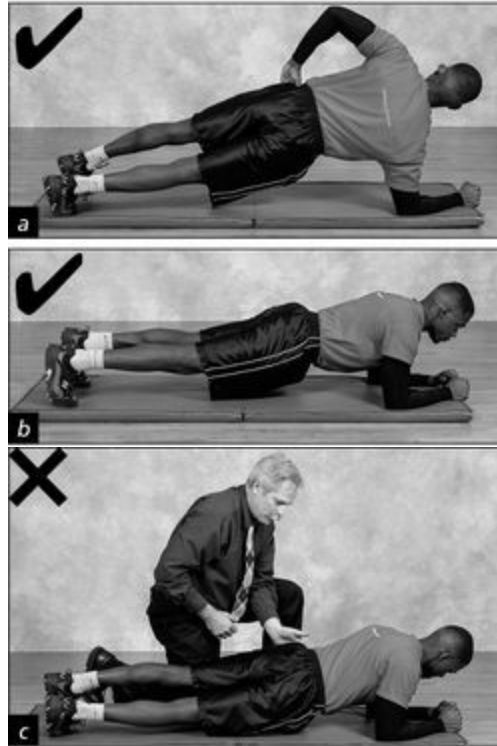


Example Cues

1. Lock the rib cage to the pelvis and focus on the latissimus dorsi initiating the turn rather than the spine twisting the muscles.
2. Roll back and forth over the elbow to train the multiple neuromuscular compartments in the abdominal wall.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Side Bridge, Advanced



Example Cues

1. Use all previous cues plus focus more on locking the rib cage to the pelvis.
2. Note that the spine twist in c is poor form.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Bird Dog, Remedial



Example Cues

1. Tune the initial spine curve to eliminate pain triggers.
2. Poor starting posture (a) is corrected (b, c).

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Bird Dog, Beginner



Focus on modifying the posture and motion to eliminate pain.

Example Cues

1. Starting posture: Restore a neutral curve to the spine.
2. Position the knees under the hips and the hands under the shoulders.
3. Tune the level of bracing to reduce or eliminate pain.
4. Lift just one hand.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Bird Dog, Intermediate



Example Cues

1. Make a fist and squeeze and push the heel away.
2. Hold for 10 seconds, then sweep the floor with the hands and knees.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Bird Dog, Advanced



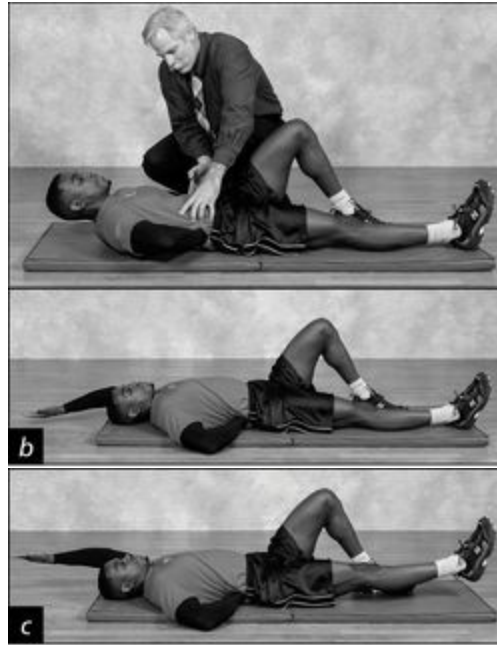


Example Cue

If your torso twists on the leg lift, reduce the height of the leg and instead push the heel away.

From S. McGill, 2016, *Low back disorders*, 3rd ed. (Champaign, IL: Human Kinetics).

Curl-Up, Highest Level



Example Cues

1. Begin with the modified curl-up, then progress to a modified dead bug.
2. Here the core is prestiffened and small, but quick motion occurs about the shoulder and opposite side hip joint.
3. No spine motion occurs.
4. Keep one hand supporting the lumbar spine.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).

Bird Dog, Highest Level







Example Cues

1. Focus on motion in the shoulder and hip joints while stiffening the spine.
2. Draw squares with one hand and one foot.

From S. McGill, 2016, Low back disorders, 3rd ed. (Champaign, IL: Human Kinetics).